SENTHIL VELAN

AI/ML Engineer | Computer Vision | Generative AI | Full-Stack & IoT

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SUMMARY

Final-year B.Tech Artificial Intelligence and Data Science student with 7.74/10 CGPA and hands-on experience building AI/ML, computer vision, Generative AI, cybersecurity, IoT, and full-stack applications. Experienced with Python, Java, TensorFlow/Keras, PyTorch, OpenCV, YOLOv8, LangChain, RAG, Flask, Django, FastAPI, SQL/NoSQL, Docker, ESP32, and cloud deployment. Builds end-to-end systems spanning model development, REST APIs, databases, dashboards, authentication, and real-time monitoring.

EDUCATION

Qualification	Institution	Duration	Score
B.Tech – Artificial Intelligence and Data Science	CARE College of Engineering, Trichy, India	2023 – 2027	CGPA: 7.74/10
Higher Secondary Education (12th)	St. James Matriculation Higher Secondary School, Trichy, India	2021 – 2023	68.67%

CORE TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, C++, SQL
- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, NLP, CNN, Transfer Learning, YOLOv8, TensorFlow/Keras, PyTorch, OpenCV, Scikit-Learn
- **Generative AI:** LLMs, Prompt Engineering, RAG, LangChain, IBM Granite, Gemini, Groq, OpenRouter, Ollama
- **Backend:** Flask, Django, FastAPI, REST APIs
- **Data / Cloud:** MySQL, PostgreSQL, MongoDB Atlas, Firebase/Firestore, Pandas, NumPy, Power BI, AWS, GCP, Azure, Supabase
- **DevOps / IoT:** Git, Docker, Gunicorn, Linux, Selenium, Figma, ESP32

EXPERIENCE

Web Development Intern — **Jun 2025 – Jul 2025**
Aerialroot Infotech
Applied web development, programming, and UI/UX fundamentals to practical application tasks.

Cloud Computing Intern — **Sep 2026 – Oct 2026**
CodeAlpha

SELECTED PROJECTS

TwinFed-OT – Federated Learning, GNN & Digital Twin Cybersecurity — *Cybersecurity / Federated Learning*

- Designed a privacy-preserving cybersecurity research platform combining Federated Learning, Graph Neural Networks, Digital Twin simulation, and SBOM/vulnerability analysis for collaborative OT threat detection and safer patch validation.
- Focused on distributed security intelligence while keeping sensitive operational data local to participating nodes.

ISL Recognition System – YOLOv8 Computer Vision — *Computer Vision / Object Detection*

- Built a real-time Indian Sign Language recognition system using YOLOv8, OpenCV, and deep learning with custom model training and fine-tuning.
- Developed a low-latency gesture detection pipeline aimed at practical accessibility and human-computer interaction.

FireGuard IoT – Real-Time Fire Monitoring & Alert System — *IoT / ESP32 / Flask*

- Built an ESP32-to-Flask IoT pipeline for sensor data ingestion, live monitoring, REST APIs, history tracking, and alert handling.
- Developed a browser-based dashboard for real-time readings and event visualization.

AI Assistant – Gemini, Flask & MongoDB Atlas — *Generative AI / Full-Stack*

- Developed a dual-layer conversational assistant combining a deterministic FAQ engine with Google Gemini for open-ended responses.
- Integrated MongoDB Atlas chat history, voice input, Docker containerization, REST endpoints, health checks, and Google Cloud Run deployment.

CERTIFICATIONS

IBM SkillsBuild: Getting Started with AI, LLM Basics, Prompt Engineering, AI Document Retrieval, RAG with LangChain • HackerRank: Python (Basic), Problem Solving (Basic), Java • Coursera: Computer Vision – YOLOv8 • Infosys Springboard: Artificial Intelligence, Cyber Security • Udemy: Machine Learning